

FoundationStereo: Zero-Shot Stereo Matching

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Presented by: Danil Smirnov, Federico Cagnolatto • 3D Data Processing Poster Session • University of Padova, 2026

Motivation

Many deep stereo matching models still rely on **per-domain fine-tuning** to achieve competitive performance.

- **Sim-to-real gap:** Synthetic training fails on real scenes
- **Challenging structures:** Reflections, textureless, thin objects
- **Weak zero-shot:** Poor generalization to unseen domains

Key Contributions

- 1 **FoundationStereo:** Stereo foundation model with strong zero-shot generalization
- 2 **FSD Dataset:** 1M synthetic stereo pairs with high diversity
- 3 **Side-Tuning Adapter:** Adapts DepthAnythingV2 for stereo matching
- 4 **Attentive Hybrid Cost Filtering:** Spatial + disparity reasoning

FSD Dataset

1 million stereo pairs via NVIDIA Omniverse with high realism.

1280×720

Resolution

1M

Pairs

Scenes: Indoor, outdoor, driving, flying

Diversity: Baselines, intrinsics, lighting

Self-curation: Removes ambiguous synthetic pairs



Figure 4: FSD dataset and self-curation

Architecture Overview

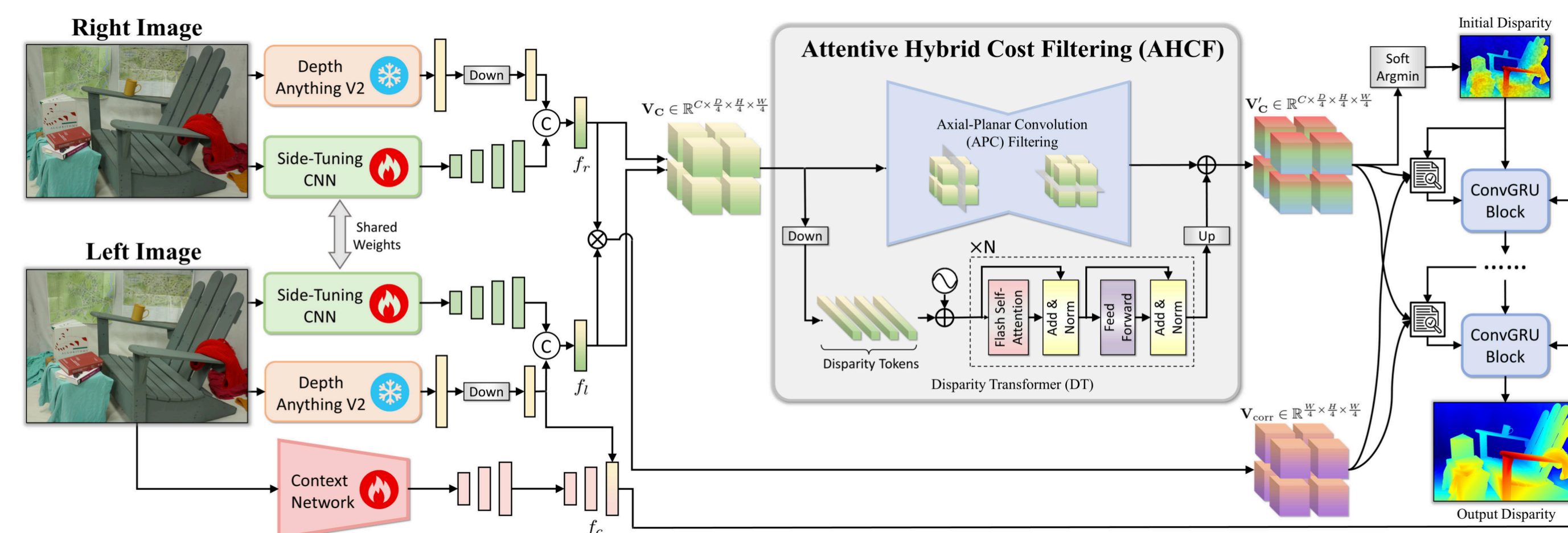


Figure 2: FoundationStereo architecture

Step 1

Feature Extraction

Step 2

Hybrid Cost Volume

Step 3

Cost Filtering

Step 4

GRU Refinement

Zero-Shot Results

Strong zero-shot performance without target-domain fine-tuning.

Dataset	Metric ↓	FoundationStereo	CREStereo++	Selective IGEV
Middlebury	BP-2	1.1	14.8	7.5
ETH3D	BP-1	0.5	4.4	3.4
KITTI-12	D1	2.3	4.7	3.2
KITTI-15	D1	2.8	5.2	4.5

Lower is better. Values are error rates (%).

Key Ideas

Side-Tuning Adapter (STA)

Adapts monocular depth features from **DepthAnythingV2** for stereo matching through lightweight adapters.

Attentive Hybrid Cost Filtering

- **Axial-Planar Conv:** Efficient spatial + disparity cost-volume filtering
- **Disparity Transformer:** Long-range reasoning over disparity space

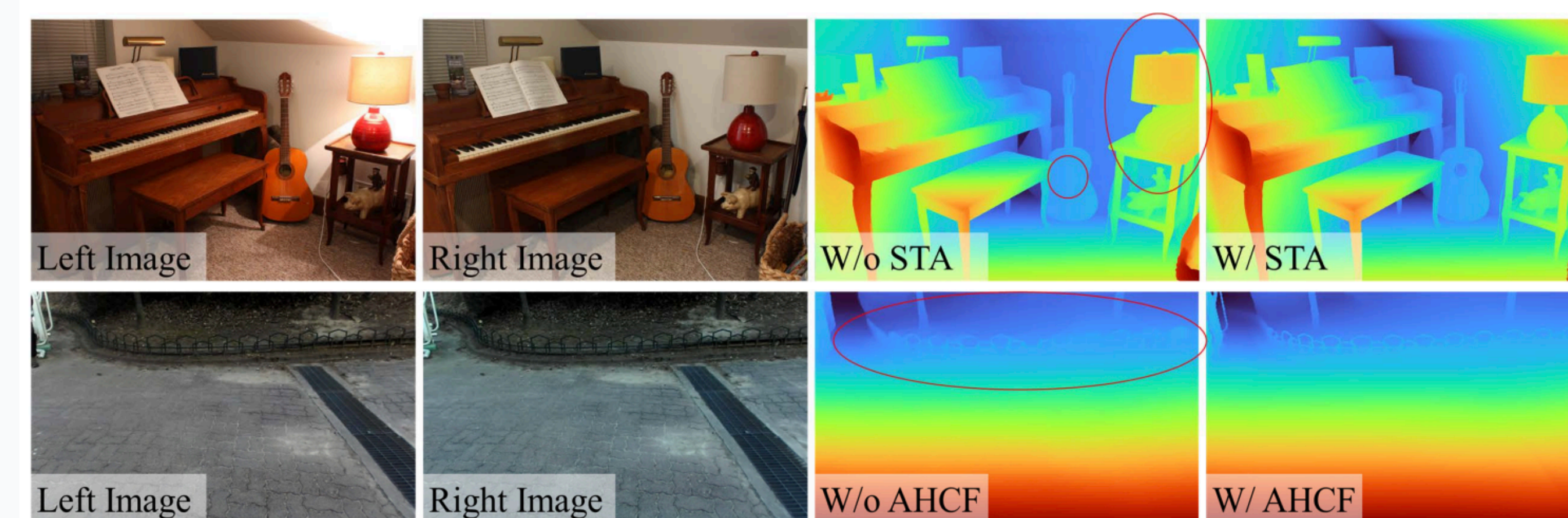


Figure 3: Effect of STA and AHCF

In-the-Wild Results

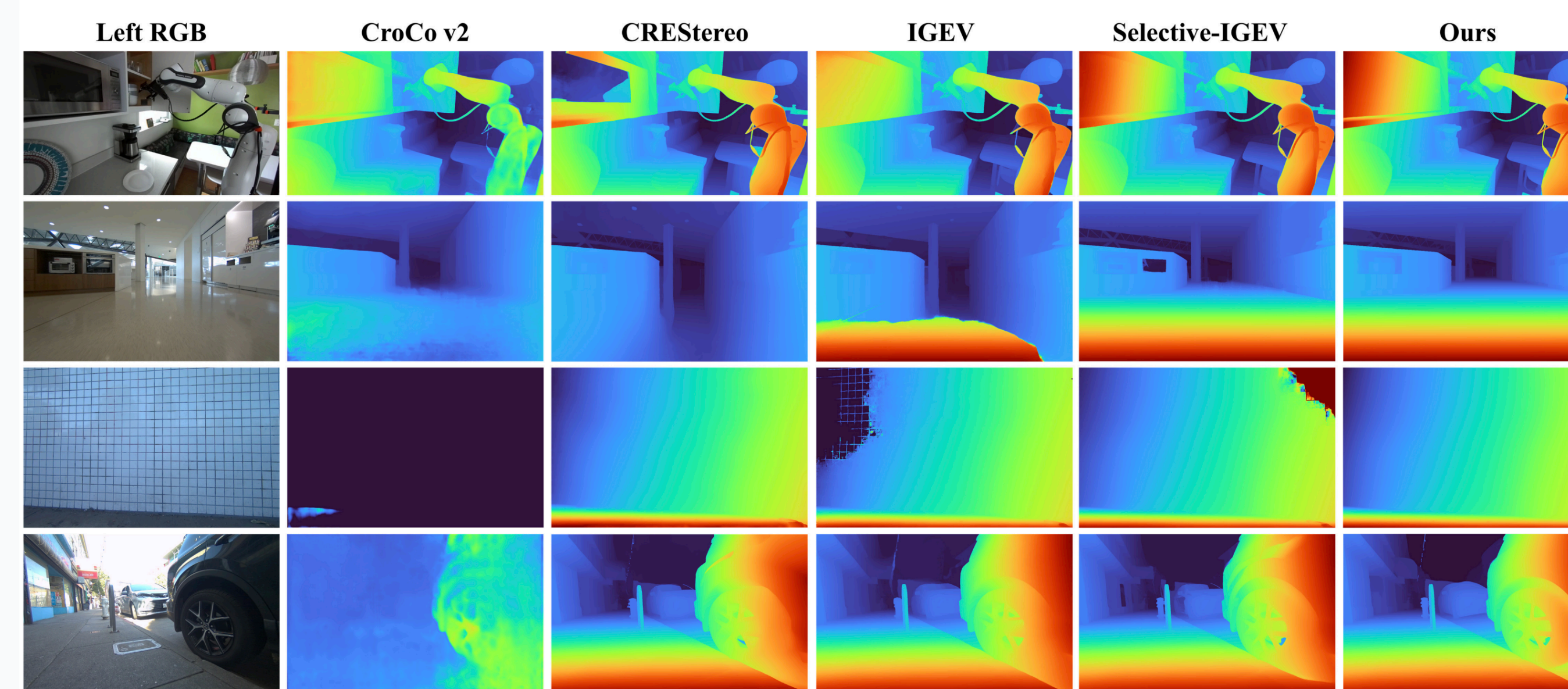


Figure 5: In-the-wild comparison

More complete disparity maps on reflective, textureless, repetitive, and thin-structure scenes.

Key Takeaways

- ✓ **Foundation model paradigm** works for stereo matching
- ✓ **Large-scale synthetic data** + monocular priors enable zero-shot transfer
- ✓ **Hybrid cost filtering** improves robust disparity estimation

Limitations

- **Efficiency:** Transformer modules need optimization for real-time use
- **Transparent objects:** Limited glass/water diversity in training data